Uncertainty Propagation in Stochastic Systems via Mixture Models with Error Quantification

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System description

Consider a discrete-time stochastic process:

$$x_{t+1} = g(x_t, \varepsilon_t), \qquad x_0 \sim \mathbb{P}_{x_0},\ \varepsilon_t \sim \mathbb{P}_{\varepsilon} \tag{1}$$

where $g : \mathbb{R}^d \times \mathbb{R}^q \to \mathbb{R}^d$ continuous, possibly non-linear function.

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Example (Additive Gaussian noise)

In the case where $g(x, \varepsilon) = f(x) + \varepsilon$ and the noise distribution is $\mathbb{P}_\varepsilon = \mathcal{N}(0, \Sigma_\varepsilon)$, we have that

$$\mathcal{K}(x, .) = \mathcal{N}(f(x), \Sigma_\varepsilon)$$

Propagating continuous distributions through non-linear functions is generally intractable

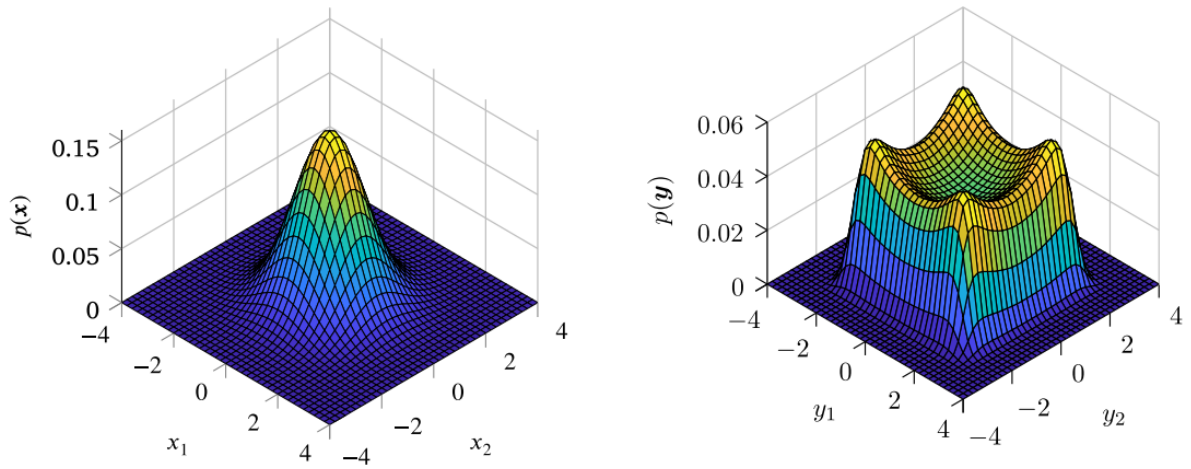
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(a) Gaussian distributed random variable.

(b) Nonlinearly transformed random variable.

Figure: Example of non-linear propagation, with $f(x) = 2\tan(x) + Ax$, and $x \sim \mathcal{N}(0, I)$ (more details in [2])

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A quick intermezzo: defining Total Variation

Definition (Total Variation)

Let $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ be a measurable space, and $\mathbb{P}, \mathbb{Q} \in \mathcal{P}(\mathcal{X})$ be probability distributions. Then, the Total Variation distance between $\mathbb{P}$ and $\mathbb{Q}$ is defined as:

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Theorem

When probability densities exist, we have:

$$TV(\mathbb{P}, \mathbb{Q}) = \frac{1}{2} \int_{\mathcal{X}} |p(x) - q(x)| dx$$

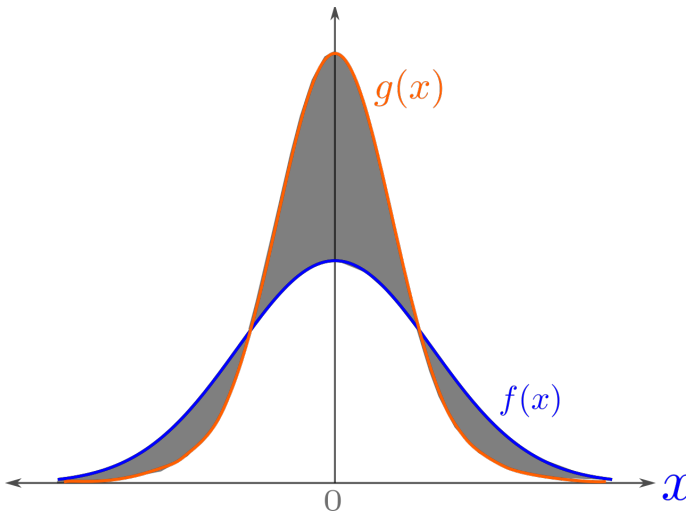


Figure: Credits: Wikipedia/MIT EDX

Problem Formulation

Problem

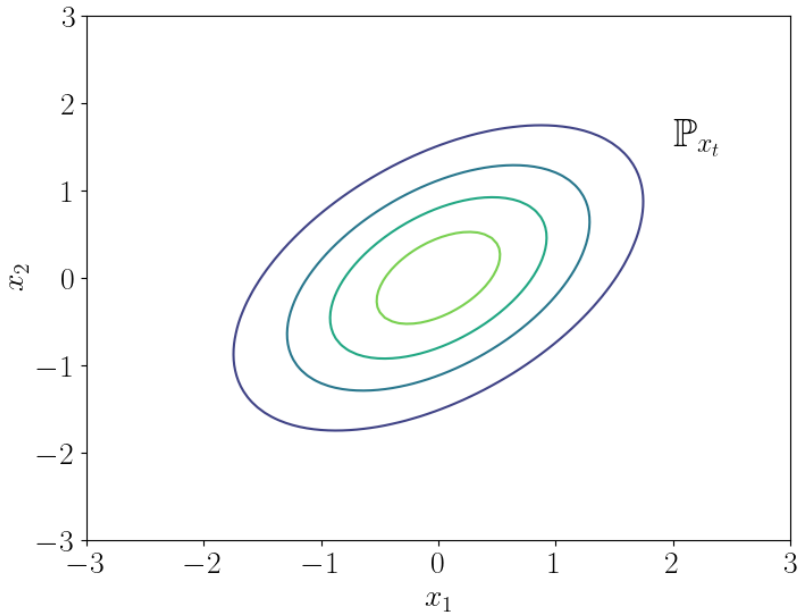
Consider a finite time horizon $\{1, ..., T\} \subset \mathbb{N}$*, and a threshold* $\delta > 0$*. Find a set of mixture distributions of finite size* $\hat{\mathbb{P}}_{x_1}, ..., \hat{\mathbb{P}}_{x_T}$ *such that for all* $t \in \{1, ..., T\}$ *it holds that:*

$$TV\left(\mathbb{P}_{x_t}, \hat{\mathbb{P}}_{x_t}\right) \leq \delta. \tag{3}$$

Stochastic Approximation Scheme

Intuitive idea:

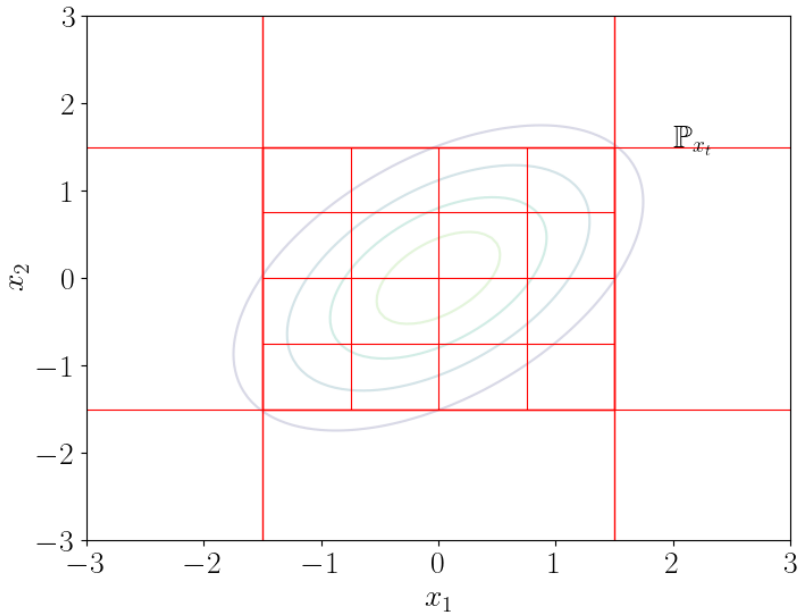
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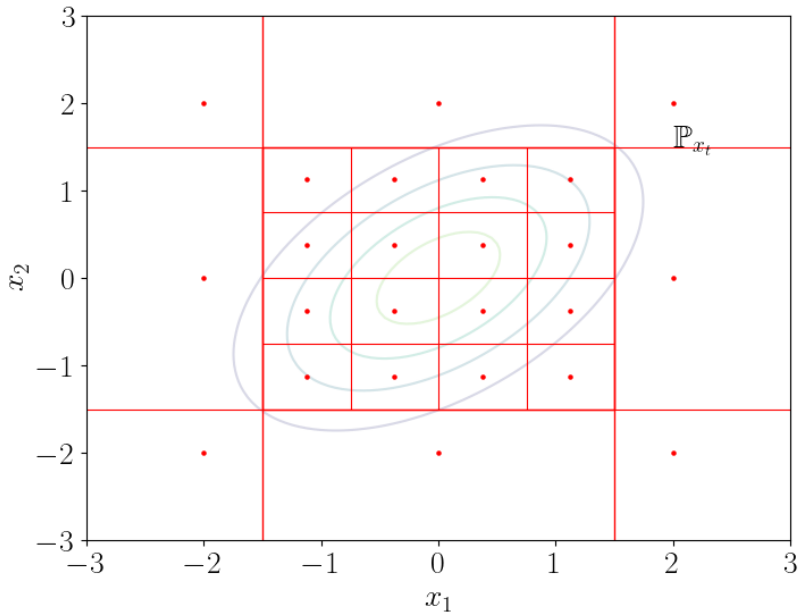
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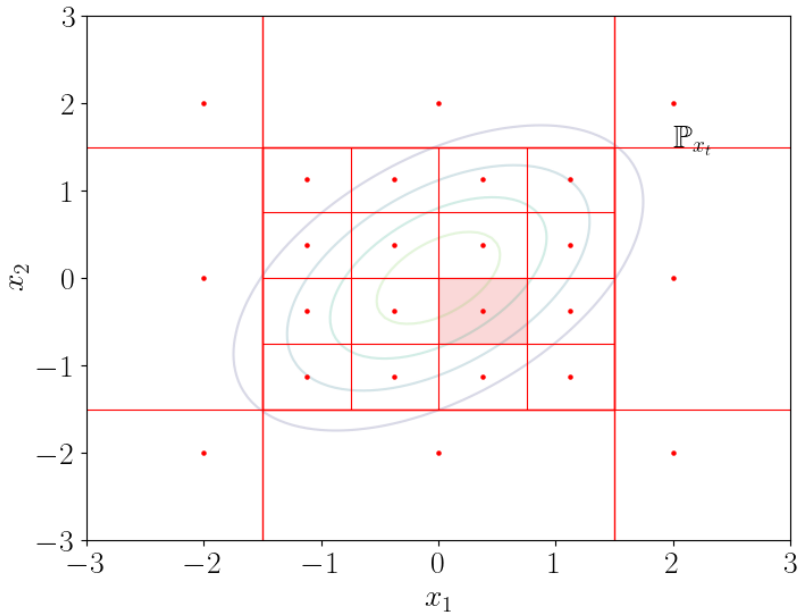
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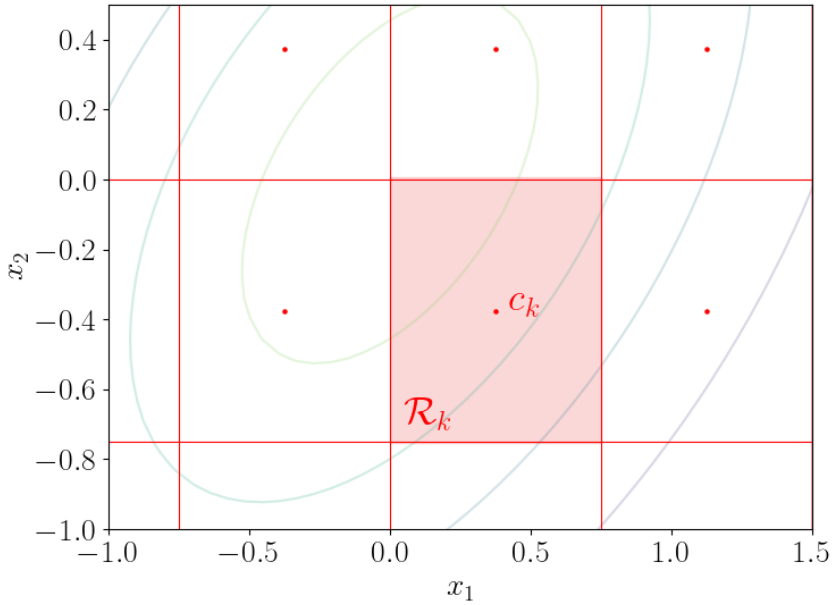
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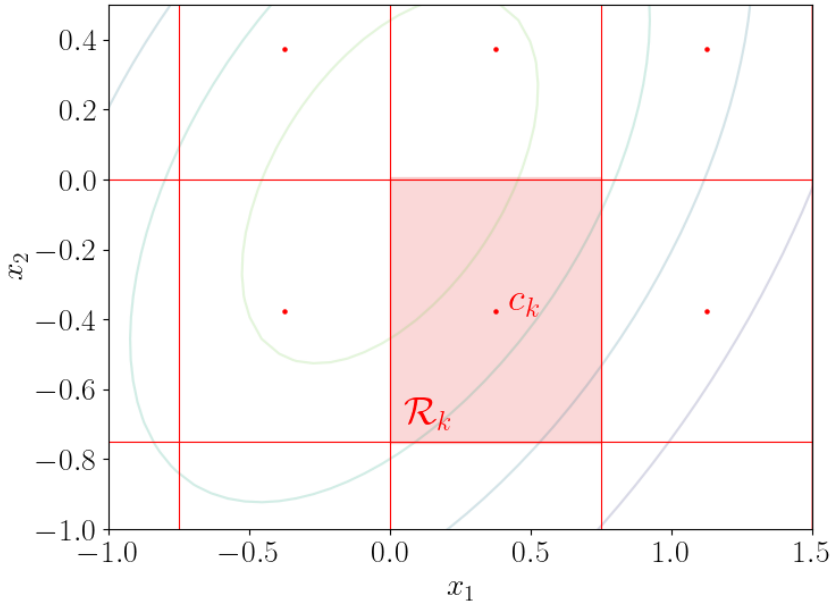
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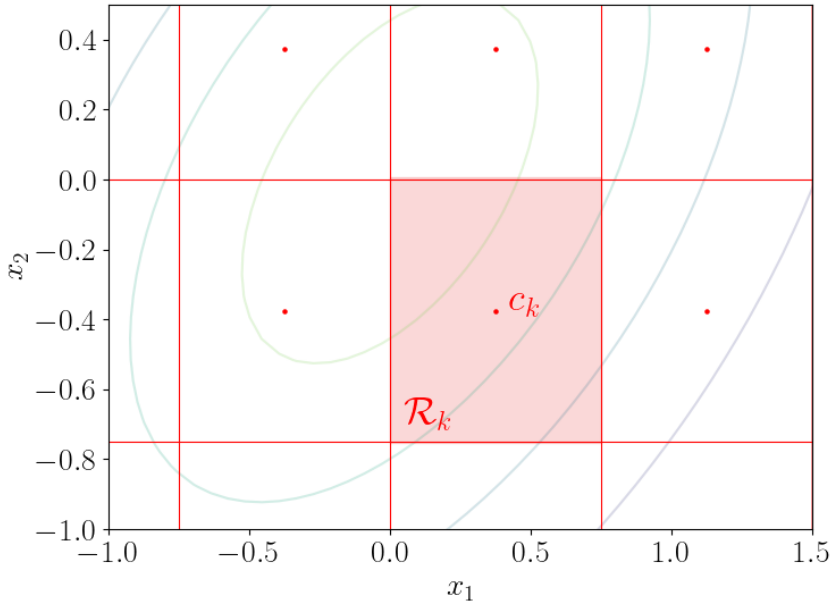
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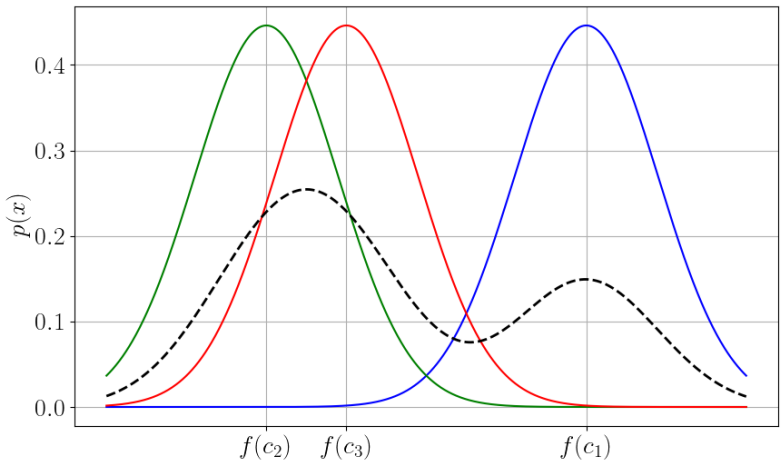
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Example ($x_{t+1} = f(x_t) + \epsilon_t$, $\epsilon_t \sim \mathcal{N}(0, \Sigma_\epsilon)$)

In this case, $\mathcal{K}(x,\cdot) = \mathcal{N}(f(x), \Sigma_\epsilon)$.

$$
\hat{\mathbb{P}}_{x_{t+1}} = \sum_{k=1}^{K} \omega_k \mathcal{N}(f(c_k), \Sigma_\varepsilon)
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Formal Guarantees in TV

Theorem

Let function $s\big(x, c_k\big)$ *be defined as*

$$s\big(x, c_k\big) := TV\big(\mathcal{K}(x, \cdot), \mathcal{K}(c_k, \cdot)\big) \tag{4}$$

Then, for any $t > 0$ *it holds that*

$$TV\big(\mathbb{P}_{x_{t+1}}, \hat{\mathbb{P}}_{x_{t+1}}\big) \leq TV\big(\mathbb{P}_{x_t}, \hat{\mathbb{P}}_{x_t}\big) + \sum_{k=1}^{K} \max_{x \in \mathcal{R}_k} s\big(x, c_k\big) \hat{\mathbb{P}}_{x_t}(\mathcal{R}_k), \tag{5}$$

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Remark: $s(x, c_k)$ can be computed in closed form for a large class of distributions, such as Gaussian

Refining the grid for better bounds in TV

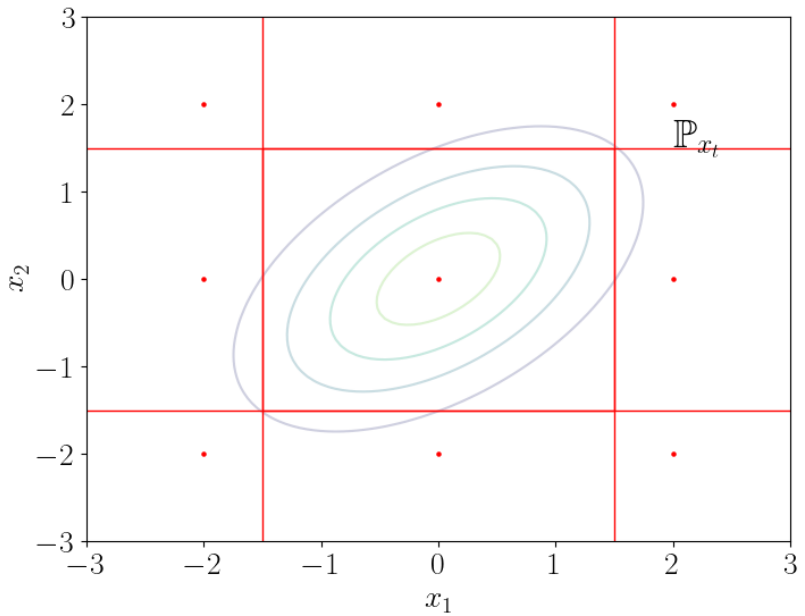
Each region contributes **independently** to the TV bound:

$$\sum_{k=1}^{K} \max_{x \in \mathcal{R}_k} s\big(x, c_k\big) \hat{\mathbb{P}}_{x_t}(\mathcal{R}_k) \qquad (6)$$

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$$s\big(x, c_k\big) = \mathrm{erf}\big(h(x, c_k)\big)$$

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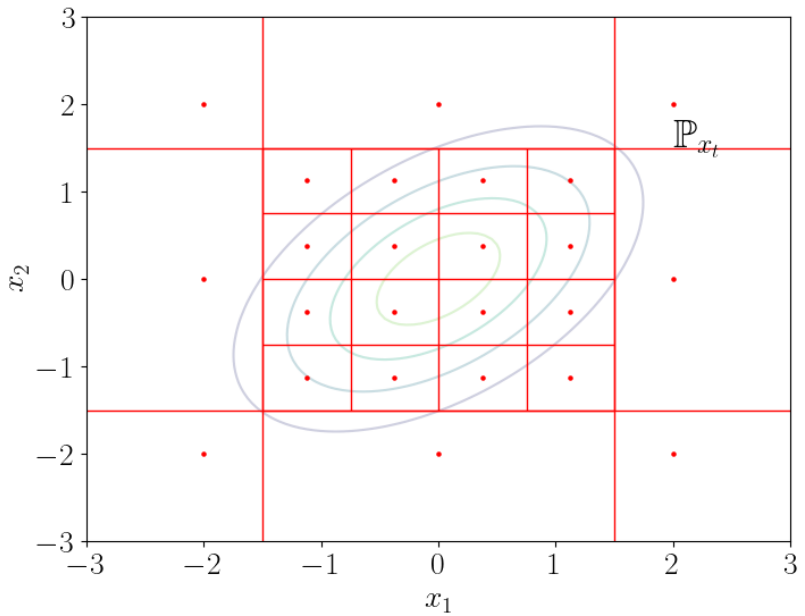
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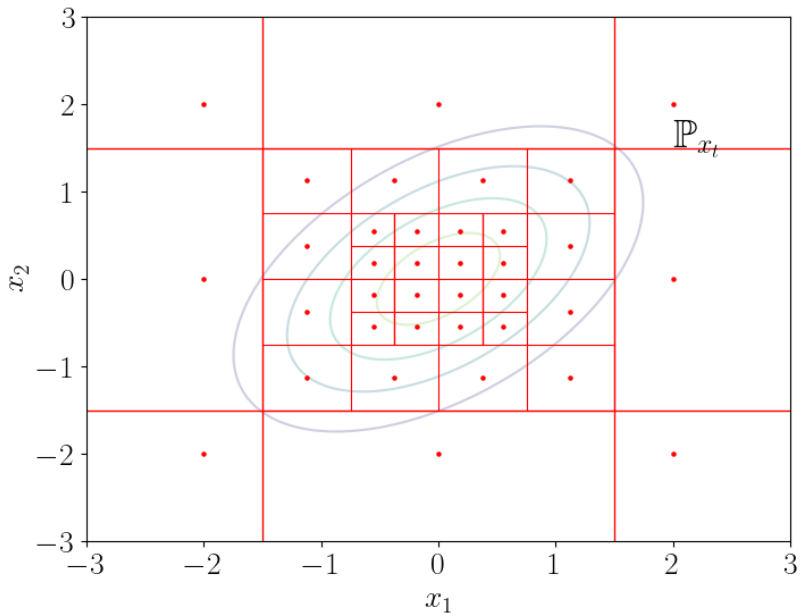
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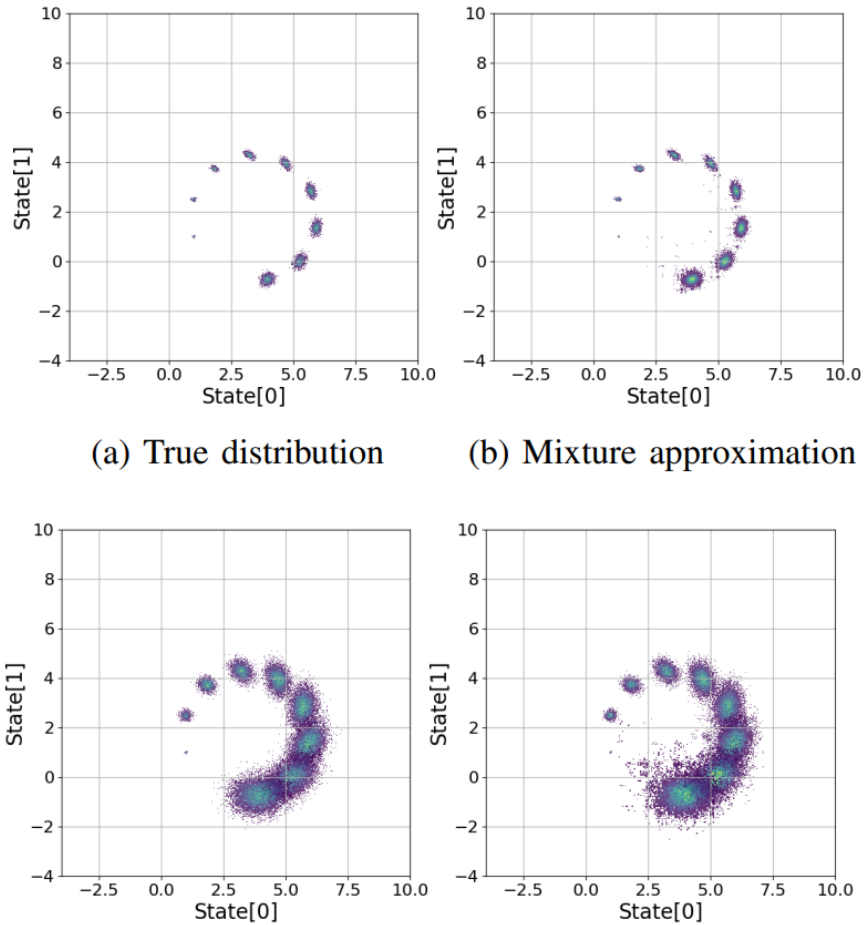
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Experimental Results

Discussion 1: Visualizing the Mixture Approximations

Dubins dynamics: $f(x) = [x_1 + v\sin(x_3), x_2 + v\cos(x_3), x_3 + u]$



(a) True distribution (b) Mixture approximation

Figure: True distribution (left) vs GMM approximation (right)

Experimental Results

Discussion 2: TV bounds for selected benchmarks

	Bimodal		Polynomial		Uniform		Dubins	
Grid method	**Alg 2**	**Equid**	**Alg 2**	**Equid**	**Alg 2**	**Equid**	**Alg 2**	**Equid**
TV_1	0.004	0.007	0.004	0.006	0.004	0.003	0.028	0.99
TV_T	0.061	0.092	0.099	0.179	0.041	0.048	0.198	1.00
Avg TV	0.033	0.049	0.039	0.070	0.022	0.022	0.101	1.00

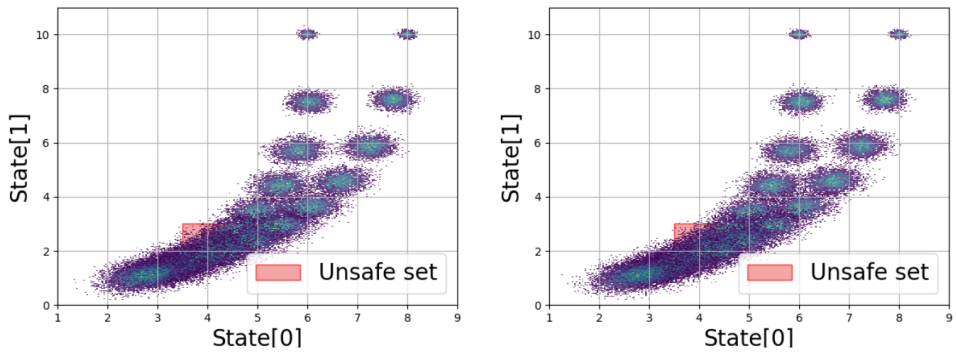
Figure: TV bounds for the first and last propagation steps (as well as for the average) of several benchmarks

Experimental Results

Discussion 3: Application to chance constrained certification

	Bimodal	
Time step	**Actual hitting probabity p_{hit}**	**Upper bound p_{hit}**
1	0.0	0.4
2	0.0	0.9
3	0.0	1.6
4	0.0	2.4
5	18.4	21.5
6	38.0	42.1
7	20.9	25.6
8	3.5	8.5
9	0.0	5.8
10	0.0	6.2

Figure: Actual p_{hit} vs probability GMM unsafe set + TV bound.



(a) True distribution (b) Mixture approximation

Experimental Results

Discussion 4: Mixture size

Unstable polynomial system: $f(x) = [f_1(x), f_2(x)]$ with

$$f_1(x) = x_1 + 1.25hx_2$$
$$f_2(x) = 1.4x_2 + 0.3h(0.25x_1^2 - 0.4x_1x_2 + 0.25x_2^2)$$

Noise variance σ^2	1		10^{-1}		10^{-2}		10^{-3}	
# refinements	**TV**	**GMM size**	**TV**	**GMM size**	**TV**	**GMM size**	**TV**	**GMM size**
Initial grid	0.020	121	0.061	124	0.205	115	0.471	127
1	0.011	485	0.031	497	0.107	461	0.279	509
2	0.006	1,937	0.016	1,982	0.054	1,841	0.151	2,033
3	0.003	7,667	0.008	7,868	0.027	7,331	0.078	8,108
4	0.002	26,288	0.005	30,824	0.014	28,997	0.039	32,213
5	0.002	31,070	0.003	59,660	0.007	112,994	0.020	127,220

Figure: TV bounds for a 1-time step propagation of the polynomial system

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- How to extend these formal guarantees for the **Wasserstein distance**?

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- How to make the method **more scalable**?
  - **Mixture compression**: How to reduce the number of components in the mixture w/ formal guarantees?
- How to extend these formal guarantees for the **Wasserstein distance**?
  - Work in progress

Thank you!

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https://github.com/EduardoFMDCosta/DUQviaTotalVariation